# Homework 09

Haoran Xu

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## Question 1 (Choosing an SVM kernel)

### Solution

**Setup.** The SAheart data have $n = 462$ rows, the binary response `chd` (160 cases, 34.6%), and nine predictors; `famhist` is recoded 1 = Absent, 2 = Present. I draw one random partition into five folds of 92 or 93 observations (seed 432) and reuse it for all candidates. Inside each fold the predictors are centred and scaled with the mean and standard deviation of the *fitting* 4/5 only, and the same transformation is applied to the held-out fifth, so no held-out information enters the fit. All fits use `e1071::svm` with `scale = FALSE` because the scaling is done by hand.

**Candidates (19 in total).** Linear kernel with $C \in \{0.01, 0.1, 1, 10\}$ (4); polynomial kernels $K(u, v) = (\gamma u^\top v + c_0)^d$ with $d \in \{2, 3, 4\}$ and $C \in \{0.01, 0.1, 1\}$ (9), holding fixed $\gamma = 1/p = 1/9$ and $c_0 = 1$ so that the polynomial contains all lower-order terms and its entries stay on a sensible scale; radial kernel $K(u, v) = \exp(-\gamma\|u - v\|^2)$ with $\gamma \in \{0.01, 0.1, 1\}$ and $C \in \{1, 10\}$ (6). The radial grid is centred near the default $\gamma = 1/p$ and the polynomial costs are lower than the linear ones because the polynomial features have a much larger range. A pilot run with radial $C = 0.1$ predicted the majority class in every fold (error 0.346), so that cost was replaced by $C = 10$.

```
suppressPackageStartupMessages(library(e1071))
d <- read.csv("data/saheart.csv")
d$famhist <- ifelse(d$famhist == "Present", 2, 1)
X <- as.matrix(d[, setdiff(names(d), "chd")]); y <- factor(d$chd)
n <- nrow(d); K <- 5
set.seed(432)
fold <- sample(rep(1:K, length.out = n))          # same five folds for every candidate

grid <- rbind(
  data.frame(kernel = "linear",     degree = NA, gamma = NA, cost = c(0.01, 0.1, 1, 10)),
  data.frame(kernel = "polynomial", degree = 2,  gamma = NA, cost = c(0.01, 0.1, 1)),
  data.frame(kernel = "polynomial", degree = 3,  gamma = NA, cost = c(0.01, 0.1, 1)),
  data.frame(kernel = "polynomial", degree = 4,  gamma = NA, cost = c(0.01, 0.1, 1)),
  expand.grid(kernel = "radial", degree = NA, gamma = c(0.01, 0.1, 1), cost = c(1, 10),
              stringsAsFactors = FALSE)[, c("kernel", "degree", "gamma", "cost")])
grid$kernel <- as.character(grid$kernel); stopifnot(nrow(grid) <= 20)

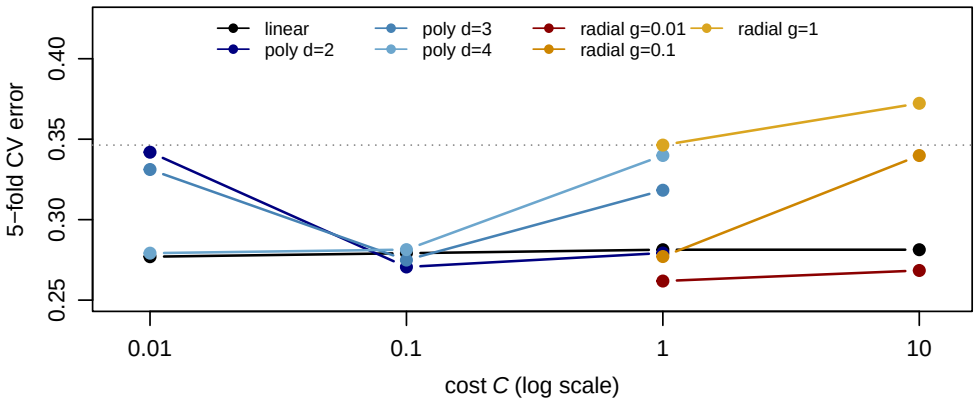
cv_err <- matrix(NA, nrow(grid), K)
for (k in 1:K) {
  tr <- fold != k
  mu <- colMeans(X[tr, ]); s <- apply(X[tr, ], 2, sd)   # fitting-fold statistics only
  Xtr <- scale(X[tr, ], mu, s); Xte <- scale(X[!tr, ], mu, s)
  for (j in seq_len(nrow(grid))) {
    g <- grid[j, ]
    fit <- switch(g$kernel,
      linear     = svm(Xtr, y[tr], kernel = "linear", cost = g$cost, scale = FALSE),
      polynomial = svm(Xtr, y[tr], kernel = "polynomial", degree = g$degree,
                       gamma = 1 / ncol(X), coef0 = 1, cost = g$cost, scale = FALSE),
```

```
      radial     = svm(Xtr, y[tr], kernel = "radial", gamma = g$gamma, cost = g$cost,
                       scale = FALSE))
    cv_err[j, k] <- mean(predict(fit, Xte) != y[!tr])
  }
}
grid$cv <- rowMeans(cv_err); grid$se <- apply(cv_err, 1, sd) / sqrt(K)
```

```
tab <- grid
tab$kernel <- ifelse(tab$kernel == "polynomial", paste0("poly d=", tab$degree), tab$kernel)
tab$gamma  <- ifelse(is.na(tab$gamma), "1/9", format(tab$gamma))
tab$gamma[tab$kernel == "linear"] <- "--"
tab <- tab[, c("kernel", "gamma", "cost", "cv", "se")]
tab$cv <- round(tab$cv, 3); tab$se <- round(tab$se, 3)
knitr::kable(tab, row.names = TRUE, align = "lrrrr",
             col.names = c("kernel", "$\\gamma$", "$C$", "mean CV error", "SE"))
```

| | kernel | $\gamma$ | $C$ | mean CV error | SE |
|---|---|---|---|---|---|
| 1 | linear | – | 0.01 | 0.277 | 0.015 |
| 2 | linear | – | 0.10 | 0.279 | 0.015 |
| 3 | linear | – | 1.00 | 0.281 | 0.016 |
| 4 | linear | – | 10.00 | 0.281 | 0.016 |
| 5 | poly d=2 | 1/9 | 0.01 | 0.342 | 0.022 |
| 6 | poly d=2 | 1/9 | 0.10 | 0.271 | 0.013 |
| 7 | poly d=2 | 1/9 | 1.00 | 0.279 | 0.014 |
| 8 | poly d=3 | 1/9 | 0.01 | 0.331 | 0.013 |
| 9 | poly d=3 | 1/9 | 0.10 | 0.275 | 0.014 |
| 10 | poly d=3 | 1/9 | 1.00 | 0.318 | 0.016 |
| 11 | poly d=4 | 1/9 | 0.01 | 0.279 | 0.010 |
| 12 | poly d=4 | 1/9 | 0.10 | 0.281 | 0.011 |
| 13 | poly d=4 | 1/9 | 1.00 | 0.340 | 0.021 |
| 14 | radial | 0.01 | 1.00 | 0.262 | 0.015 |
| 15 | radial | 0.10 | 1.00 | 0.277 | 0.014 |
| 16 | radial | 1.00 | 1.00 | 0.346 | 0.012 |
| 17 | radial | 0.01 | 10.00 | 0.268 | 0.017 |
| 18 | radial | 0.10 | 10.00 | 0.340 | 0.019 |
| 19 | radial | 1.00 | 10.00 | 0.372 | 0.014 |

```
par(mar = c(4, 4, 1, 1), mgp = c(2.3, 0.8, 0), cex = 0.85)
plot(NA, xlim = log10(c(0.008, 12)), ylim = c(0.25, 0.425), xaxt = "n",
     xlab = expression(cost ~ italic(C) ~ "(log scale)"), ylab = "5-fold CV error")
axis(1, at = log10(c(0.01, 0.1, 1, 10)), labels = c(0.01, 0.1, 1, 10))
abline(h = mean(y == 1), lty = 3, col = "grey50")           # always predict 0
grp <- with(grid, ifelse(kernel == "radial", paste0("radial g=", gamma),
                  ifelse(kernel == "polynomial", paste0("poly d=", degree), "linear")))
cols <- c("linear" = "black", "poly d=2" = "navy", "poly d=3" = "steelblue",
          "poly d=4" = "skyblue3", "radial g=0.01" = "darkred",
          "radial g=0.1" = "orange3", "radial g=1" = "goldenrod")
for (g in names(cols)) { i <- grp == g
  lines(log10(grid$cost[i]), grid$cv[i], type = "b", pch = 19, col = cols[g], lwd = 1.5) }
legend("top", legend = names(cols), col = cols, pch = 19, lty = 1, bty = "n",
       cex = 0.75, ncol = 4)
```




**Reading the table.** The dotted line in the figure is the error of always predicting "no CHD" (0.346); the radial kernel with $\gamma = 1$ never leaves it, and neither do the polynomial kernels at their smallest cost, so those settings are under-fitted. The seven best candidates all sit between 0.26 and 0.28: radial $\gamma = 0.01$ with $C = 1$ (0.262) and $C = 10$ (0.268), quadratic $C = 0.1$ (0.271), cubic $C = 0.1$ (0.275), and every linear cost (0.277 to 0.281). The linear kernel is remarkably flat in $C$: once the margin is set, changing the cost by three orders of magnitude moves the error by 0.004. The fold-to-fold standard errors are about 0.015, so the gap between the overall minimum and the linear kernel (0.015) is one standard error, and the gap to the quadratic kernel is less than that.

**Preference.** I prefer the **radial kernel with $\gamma = 0.01$ and $C = 1$**, the candidate with the lowest cross-validated error, but with a caveat. A radial kernel with $\gamma$ this small is almost linear on standardized predictors ($\|u - v\|^2$ is at most a few tens, so $\exp(-0.01\|u - v\|^2)$ is close to $1 - 0.01\|u - v\|^2$), which is why it and the linear kernel agree so closely and why the higher-$\gamma$ radial settings, which can bend the boundary, do worse. The evidence therefore says the SAheart boundary is essentially linear in the nine standardized predictors: flexible kernels (degree 3 and 4, radial $\gamma \geq 0.1$) never beat the linear fit and over-fit quickly as $C$ grows. If interpretability mattered more than the last 0.01 of error I would report the linear SVM with $C = 0.01$ instead, since its error is within one SE of the minimum and it exposes a single coefficient vector; on pure predictive grounds the near-linear radial fit is the choice.

## Question 2 (Write and use a short SVM analysis skill)

### Solution

**Writing the skill.** The skill is `svm-skill` (full text in Appendix B). It was written by hand, informed by two sources of feedback: the review I received on my Homework 7 skill (Issue 3: hard-code the default grid step into the skill, and state whether the final pick is the empirical CV minimum or a one-SE parsimony choice) and the issues I found when reviewing a classmate's skill in Homework 8 (prose and report drawn from different runs, a greedy minimum-CV rule across many noisy candidates, an unresolved "one table" rule versus a confusion matrix, undefined coding of text predictors). The body of the skill is under 50 lines and names no dataset, variable, or result.

**Applying the skill.** The skill was invoked by name through a background Claude Code agent, with only the data facts supplied in the prompt (file path, response `chd`, the nine predictor names, `famhist` coded 1 = Absent, 2 = Present, seed 432) plus the file names of the deliverables. The agent was told not to read Question 1 or any other file in the folder. The prompt, abridged only by shortening paths:

```
Use the skill `svm-skill`: read ~/.claude/skills/svm-skill/SKILL.md first and
follow it exactly, section by section. Do NOT edit SKILL.md.
Data: homework/week-09/data/saheart.csv (462 rows, header). Response: `chd`
(0/1). Predictors: the other nine columns: sbp, tobacco, ldl, adiposity,
famhist, typea, obesity, alcohol, age. `famhist` coded 1 = "Absent",
2 = "Present". Random seed: 432 (set once before the test split). Language: R
with e1071::svm (scale = FALSE, standardize yourself as the skill says),
knitr::kable, base graphics.
```

```
Deliverables: q2-analysis.R (reproduces everything from the seed; saves all
results to q2-results.rds and the CV table to q2-cv-table.csv); q2-report.qmd
(Quarto fragment: no YAML, echo: false, level-4 headings Data, Candidates,
Selection, Result, Limitations; every number read from q2-results.rds by
inline R, never retyped; figure 5 x 2.4 in, no caption; <= 300 words);
q2-skill-notes.md (the separate note on ambiguous instructions).
```

**Checking the result.** I re-ran `q2-analysis.R` from the seed and obtained an identical results file, so every number in Appendix A comes from one run and is read from `q2-results.rds` by inline code rather than retyped. In the script I verified that the 25% stratified test set (116 rows: 76 controls, 40 cases) is drawn before anything else and touched only once; that the five stratified folds (24 cases in each) are built once and reused for all 16 settings; that the centring and scaling inside each fold use only the fitting rows, and the final refit standardizes the test rows with training means and SDs; and that the grid, the one-SE rule, and the simplicity order are exactly those the skill prescribes. I then recomputed the selection from the saved CV table: the minimum is the quadratic kernel with $C = 1$ (CV error 0.254, SE 0.019), the one-SE threshold is 0.273, and the nearest simpler setting (radial $\gamma = 0.5/p$, $C = 1$, error 0.275) misses it, so the pick is the minimum itself. The reported test error $32/116 = 0.276$ agrees with the confusion matrix.

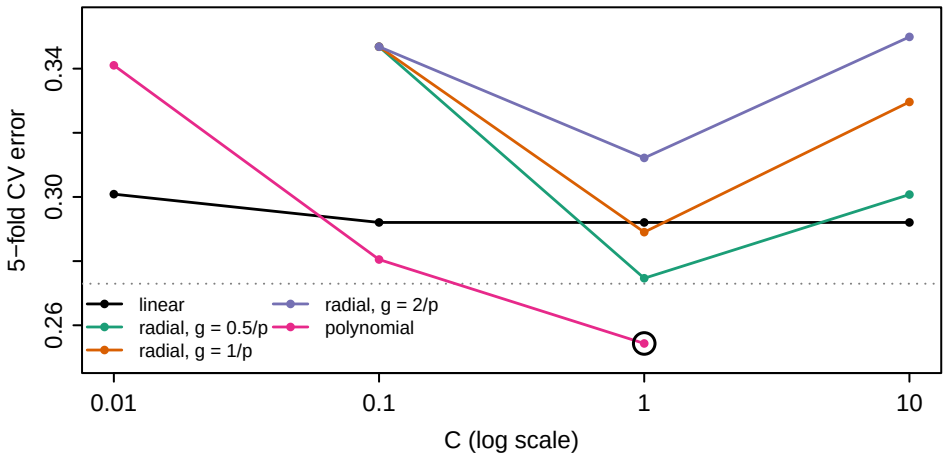
**Agreement with Question 1.** The skill chose the quadratic kernel, whereas Question 1 preferred a near-linear radial kernel ($\gamma = 0.01$) with the quadratic and linear fits within one SE. The two runs are not contradictory. The skill tunes on 346 training rows with a different fold split, its radial grid starts at $\gamma = 0.5/p \approx 0.056$ and so never contains the almost-linear $\gamma = 0.01$ that won in Question 1, and in both analyses the spread between the best curved kernel and the linear kernel (here 0.254 versus 0.292) is only about two fold-to-fold SEs. Both reach the same substantive conclusion: the boundary is close to linear with at most mild curvature, and the held-out test error of 0.276 is in line with the 0.26 to 0.28 cross-validated errors of Question 1.

**Ambiguities the agent reported** (18 in total, in `q2-skill-notes.md`). The ones I would fix in the skill: (i) at run time the description still said to trigger on the skill's draft name rather than `svm-skill`, so the agent took the explicit file path as the trigger (fixed since); (ii) "within one SE of the minimum" does not say whose SE, so the agent used the SE of the minimum setting; (iii) "whether the winning margin exceeds one SE" is undefined, and was read as the best polynomial setting versus the best radial one; (iv) "one table: every setting … best row marked" cannot fit one page as a 16-row table, so the agent laid the settings out as a kernel-by-$C$ matrix of "mean (SE)", which departs from the letter of the rule; (v) the one-page limit, not the 300-word limit, was binding, and the page holds only because the appendix below tightens Quarto's heading spacing; and (vi) the skill is silent on how to stratify folds, how to round a 25% split per class, and whether the binary `famhist` is standardized like the other predictors. None of these changed the conclusion, but (ii) to (iv) are silent choices a reader cannot see from the report.

**Appendix A: one-page analysis report produced by the skill**

**Data** 462 subjects, 9 predictors (`famhist`: 1 = Absent, 2 = Present), response `chd` with 160 cases (34.6%), no missing values, n/p ≈ 51. A stratified 25% test set (116 rows) was held out first; 5 stratified folds were built once on the 346 training rows, standardizing within each fold from the fitting rows.

**Candidates** Linear (baseline: flat boundary), radial (local curvature), and polynomial degree 2 (global quadratic curvature, added because n > 200). The 16 settings in the table (polynomial: $\gamma = 1/\text{p}$, coef0 = 1) were scored by 5-fold misclassification rate, SE = $\text{sd}/\sqrt{5}$.




| mean (SE); * chosen | C = 0.01 | C = 0.1 | C = 1 | C = 10 |
|---|---|---|---|---|
| linear | 0.301 (0.028) | 0.292 (0.023) | 0.292 (0.023) | 0.292 (0.023) |
| radial, g = 0.5/p | – | 0.347 (0.001) | 0.275 (0.021) | 0.301 (0.022) |
| radial, g = 1/p | – | 0.347 (0.001) | 0.289 (0.018) | 0.330 (0.018) |
| radial, g = 2/p | – | 0.347 (0.001) | 0.312 (0.021) | 0.350 (0.028) |
| polynomial | 0.341 (0.003) | 0.280 (0.016) | **0.254 (0.019)** * | – |

**Selection** Minimum 0.254 (SE 0.019) at polynomial, C = 1; one-SE threshold 0.273. The best linear (0.292) and radial (0.275) settings sit above it, so 1 setting qualifies and the rule picks polynomial, C = 1.

**Result** Refit on all 346 training rows: test error 27.6% (34.5% for always predicting no CHD). Confusion matrix: of 76 non-cases, 66 correct and 10 false positives; of 40 cases, 18 detected and 22 missed; 196 support vectors (56.6% of training rows). Its winning margin over the best radial setting, 0.020, just exceeds one SE. Quadratic beating radial and linear suggests a gently curved global boundary.

**Limitations** One split and five folds give a noisy SE, so the kernel ranking is tentative. Many support vectors signal heavy class overlap, and the error rate ignores imbalance and asymmetric costs.

**Appendix B: complete `SKILL.md`**

```
---
name: svm-skill
description: Fit and tune an SVM classifier and write a one-page report. Use only when the
 ↪ student explicitly names "svm-skill" and supplies a dataset, response, and predictors;
 ↪ never trigger automatically.
---

# SVM analysis skill

The student gives the data file, the response, the predictors, any coding of text
columns, and a random seed. Use only those inputs; never guess variable names.

## 1. Suitability
Check: binary (or few-class) response; numeric or clearly coded predictors; n at least
a few times p; no missing values (stop and report if any). Note class balance.
If the response is continuous or n < 50, say SVM is a poor fit and stop.

## 2. Kernels
Always include linear (baseline) and radial. Add polynomial degree 2 only if n > 200.
Explain in one sentence why each is included and skip degrees above 2.

## 3. Tuning parameters
Linear: C in {0.01, 0.1, 1, 10}. Radial: C in {0.1, 1, 10} x gamma in {0.5/p, 1/p, 2/p}.
Polynomial: C in {0.01, 0.1, 1}, gamma = 1/p, coef0 = 1. At most 20 settings total.
Standardize predictors inside each fold using fitting rows only.

## 4. Evaluation and selection
Reserve a 25% stratified test set first. Build 5 stratified folds once on the training
rows and reuse them for every setting. Score by misclassification rate; report mean and
SE = sd(fold rates) / sqrt(5). Select by the one-SE rule: the simplest setting (linear
before radial before polynomial, then smaller C) whose CV error is within one SE of the
minimum. Refit the choice on all training rows and evaluate on the test set once.

## 5. Figure and table
One figure: CV error vs log C, one line per kernel/gamma, y-range fitted to the data,
no caption. One table: every setting with mean CV error and SE, best row marked.
The confusion matrix goes in prose, not a second table.

## 6. Interpretation
Report: chosen setting, its CV error and SE, test error, number of support vectors,
and whether the winning margin exceeds one SE. State in one sentence what the kernel
comparison implies about the shape of the boundary.

## 7. Report format
Write a Quarto fragment (no YAML, echo: false) with headings Data, Candidates,
Selection, Result, Limitations. At most 300 words, so it fits one page at 11 pt with
the figure (5 x 2.4 in) and the table. Every number must come from one script run saved
to a results file; read numbers from that file, never retype them.
Report any instruction in this skill that was ambiguous in a separate note.
```